# Calculus of functions of multiple variables

# Solutions to Problem Sheet 3

1. (a) $\dfrac{\partial f}{\partial x} = y + \dfrac{1}{y}, \qquad \dfrac{\partial f}{\partial y} = x - \dfrac{x}{y^2}, \qquad \dfrac{\partial^2 f}{\partial x^2} = 0, \qquad \dfrac{\partial^2 f}{\partial y^2} = \dfrac{2x}{y^3},$

$$\frac{\partial^2 f}{\partial x \partial y} = 1 - \frac{1}{y^2} = \frac{\partial^2 f}{\partial y \partial x}.$$

(b) $\dfrac{\partial f}{\partial x} = 2xy^3\cos(x^2y^3), \qquad \dfrac{\partial f}{\partial y} = 3y^2x^2\cos(x^2y^3),$

$$\frac{\partial^2 f}{\partial x^2} = 2y^3\cos(x^2y^3) - 4x^2y^6\sin(x^2y^3), \quad \frac{\partial^2 f}{\partial y^2} = 6yx^2\cos(x^2y^3) - 9y^4x^4\sin(x^2y^3),$$

$$\frac{\partial^2 f}{\partial x \partial y} = 6y^2x\cos(x^2y^3) - 6x^3y^5\sin(x^2y^3) = \frac{\partial^2 f}{\partial y \partial x}.$$

2. $u = x^4 - 6x^2y^2 + y^4, \qquad v = 4x^3y - 4xy^3,$

$$\frac{\partial u}{\partial x} = 4x^3 - 12xy^2, \qquad \frac{\partial u}{\partial y} = -12x^2y + 4y^3,$$

$$\frac{\partial v}{\partial x} = 12x^2y - 4y^3, \qquad \frac{\partial v}{\partial y} = 4x^3 - 12xy^2.$$

(a) $\dfrac{\partial^2 u}{\partial x^2} = 12x^2 - 12y^2, \qquad \dfrac{\partial^2 u}{\partial y^2} = -12x^2 + 12y^2, \qquad \text{so} \quad \dfrac{\partial^2 u}{\partial x^2} + \dfrac{\partial^2 u}{\partial y^2} = 0.$

(b) $\dfrac{\partial^2 u}{\partial x^2} = \dfrac{\partial^2 v}{\partial x \partial y}, \quad \dfrac{\partial^2 u}{\partial y^2} = -\dfrac{\partial^2 v}{\partial y \partial x} \qquad \text{so} \quad \dfrac{\partial^2 u}{\partial x^2} + \dfrac{\partial^2 u}{\partial y^2} = \dfrac{\partial^2 v}{\partial x \partial y} - \dfrac{\partial^2 v}{\partial y \partial x} = 0.$

3. (a) Differentiating, we obtain $f_x = 2x - y + 2,\ f_y = -x + 2y + 2$. To find the stationary points, we solve

$$\begin{aligned} 2x - y &= -2, \\ -x + 2y &= -2. \end{aligned}$$

Hence, $2x - y = -x + 2y \Rightarrow x = y = -2$, i.e. $(-2, -2)$ is the only stationary point. To determine the nature of the stationary point, we find
$f_{xx} = 2,\ f_{yy} = 2,\ f_{xy} = -1$, hence $f_{xx}f_{yy} - f_{xy}^2 = 3 \ > \ 0$ and $f_{xx} > 0$ so f has a local minimum at $(-2, -2)$;

(b) Differentiating, we obtain $f_x = 4x + y - 11,\ f_y = x - 6y + 16$. To find the stationary points, we solve

$$\begin{aligned} 4x + y &= 11, \quad (1) \\ x - 6y &= -16. \quad (2) \end{aligned}$$

$(1) - 4(2)$ yields $25y = 75 \Rightarrow y = 3 \Rightarrow x = 2$, i.e. the only stationary point is $(2, 3)$. To determine the nature of the stationary point, we find $f_{xx} = 4,\ f_{yy} = -6,\ f_{xy} = 1$ hence $f_{xx}f_{yy} - f_{xy}^2 = -25\ < 0$ so f has a saddle point at $(2, 3)$.

(c) Differentiating, we obtain $f_x = 2x - 8x^3$, $f_y = 2y$. To find the stationary points, we solve

$$\begin{aligned} 2x - 8x^3 = 2x(1 - 4x^2) &= 0, \\ 2y &= 0. \end{aligned}$$

This system of equations has solutions $x = 0,\ \pm 1/2,\ y = 0$. Hence, the stationary points are $(0, 0)$, $(\pm 1/2, 0)$. To determine the nature of the stationary points, we find $f_{xx} = 2 - 24x^2$, $f_{yy} = 2$, $f_{xy} = 0$, hence $f_{xx}f_{yy} - f_{xy}^2 = 4 - 48x^2$. Since $y = 0$ for all three stationary points, we need only consider the three different $x$ values.

$\underline{x = 0}$

$4 - 48(0)^2 = 4 > 0$, $f_{xx} = 2 > 0$, so $(0, 0)$ is a local minimum.

$\underline{x = \pm 1/2}$

$4 - 48(\pm 1/2)^2 = 4 - 12 = -8 < 0$, so $(\pm 1/2, 0)$ are both saddle points.

4. To find the stationary points we solve

$$\begin{aligned} f_x &= 8x^3 - 6xy^2 + 16x = 2x(4x^2 - 3y^2 + 8) = 0, \\ f_y &= -6x^2y + 4y^3 + 6y = 2y(2y^2 - 3x^2 + 3) = 0. \end{aligned}$$

One solution is $x = y = 0$ To obtain more solutions, we consider the system

$$\begin{aligned} 4x^2 - 3y^2 &= -8, \quad (1) \\ 3x^2 - 2y^2 &= 3. \quad (2) \end{aligned}$$

$3(2) - 2(1)$ yields $x^2 = 25 \Rightarrow x = \pm 5$. Substituting $x = \pm 5$ into either of $(1)$ or $(2)$ gives $y = \pm 6$. Hence, the function $f$ has 5 stationary points, with coordinates $(0, 0)$, $(5, \pm 6)$, $(-5, \pm 6)$.

To determine the nature of the stationary points, we find $f_{xx} = 24x^2 - 6y^2 + 16$, $f_{yy} = -6x^2 + 12y^2 + 6$, $f_{xy} = -12xy$.

$\underline{(x, y) = (0, 0)}$

$f_{xx}f_{yy} - f_{xy}^2 = (16)(6) - 0 = 96 > 0,\ f_{xx} = 16 > 0$, so $(0, 0)$ is a local minimum.

$\underline{(x, y) = (5, 6)}$

$f_{xx}f_{yy} - f_{xy}^2 = (400)(288) - 360^2 = -14400 < 0$, so $(5, 6)$ is a saddle point.

$\underline{(x, y) = (5, \pm 6),\ (-5, \pm 6)}$

$f_{xx}f_{yy} - f_{xy}^2 = (400)(288) - 360^2 = -14400 < 0$, so $(5, 6)$, $(5, -6)$, $(-5, 6)$, $(-5, -6)$ are all saddle points.

5. To find $a$, $b$ and $c$ that minimise the sum $S = (a - 32)^2 + (b - 64)^2 + (c - 82)^2$, we first eliminate $c$ using $a + b + c = 180$, which yields

$$S = (a - 32)^2 + (b - 64)^2 + (98 - a - b)^2.$$

To minimise $S$, we must solve the system

$$S_a = 2(a-32) - 2(98-a-b) = 2(2a+b-130) = 0,$$
$$S_b = 2(b-64) - 2(98-a-b) = 2(a+2b-162) = 0,$$

i.e.

$$2a + b = 130,$$
$$a + 2b = 162.$$

This system has solutions $a = 32\frac{2}{3}$, $b = 64\frac{2}{3}$. Finally, $a+b+c = 180$ gives $c = 82\frac{2}{3}$. Hence, the values of the angles $a$, $b$ and $c$ that minimise the error are

$$a = 32\frac{2}{3}^\circ,\ b = 64\frac{2}{3}^\circ,\ c = 82\frac{2}{3}^\circ.$$

To prove that this is a minimum of $S$, we find $S_{aa} = 4$, $S_{bb} = 4$, $S_{ab} = 2$, so $S_{aa}S_{bb} - S_{ab}^2 = 12 > 0$, $S_{aa} > 0$, so the point above is indeed a minimum.

6. We deal with the constraint first. Since the point of interest lies on the plane $2x + 3y + 4z = 58$, we have $x = (58 - 3y - 4z)/2$. Let $d^2 = x^2 + y^2 + z^2$. As implied by the question, minimising $d^2$ is equivalent to minimising $d$. Given the constraint, we minimise

$$D = d^2 = (58 - 3y - 4z)^2/4 + y^2 + z^2.$$

We find $y$ and $z$ by solving the system

$$D_y = -\frac{3}{2}(58 - 3y - 4z) + 2y = 0,$$
$$D_z = -2(58 - 3y - 4z) + 2z = 0.$$

Simplifying,

$$13y + 12z = 174, \quad (1)$$
$$3y + 5z = 58. \quad (2)$$

$5(1) - 12(2)$ yields $29y = 174 \Rightarrow y = 6 \Rightarrow z = 8$. $2x + 3y + 4z = 58$ gives $x = 4$. Hence, the nearest point on the plane to the origin is $(4, 6, 8)$.

7. $\dfrac{df}{dt} = \dfrac{\partial f}{\partial x} \cdot \dfrac{dx}{dt} + \dfrac{\partial f}{\partial y} \cdot \dfrac{dy}{dt}$

(a) $f_x = 2x(1+y^2)$, $f_y = 2y(1+x^2), x_t = -\sin t$, $y_t = \cos t$, thus substituting for $x$ and $y$ in $f_x$ and $f_y$ ($x = \cos t$ and $y = \sin t$) gives

$$\begin{aligned}\frac{\mathrm{d}f}{\mathrm{d}t} &= \frac{\partial f}{\partial x}\frac{\mathrm{d}x}{\mathrm{d}t} + \frac{\partial f}{\partial y}\frac{\mathrm{d}y}{\mathrm{d}t}, \\ &= 2\cos t(1+\sin^2 t)(-\sin t) + 2\sin t(1+\cos^2 t)(\cos t), \\ &= 2\sin t\cos t(\cos^2 t - \sin^2 t).\end{aligned}$$

(b) $f = x^2 + y^2 + x^2y^2 = \cos^2 t + \sin^2 t + \cos^2 \sin^2 t = 1 + \sin^2 t\cos^2 t$. Differentiating this expression with respect to $t$ yields

$$\begin{aligned}\frac{\mathrm{d}f}{\mathrm{d}t} &= 2\sin t\cos t\cos^2 t - 2\sin t\cos t\sin^2 t, \\ &= 2\sin t\cos t(\cos^2 t - \sin^2 t).\end{aligned}$$

8. Using $x = r\cos\theta$, $y = r\sin\theta$, we obtain

$$\begin{aligned}\bar{T} &= (r\cos\theta)^3 - 3(r\cos\theta)(r^2\sin^2\theta),\\ &= r^3\cos^3\theta - 3r^3\cos\theta\sin^2\theta,\\ &= r^3\cos 3\theta,\end{aligned}$$

using the given identity. Using the change of variables formula, we obtain

$$\frac{\partial\bar{T}}{\partial r} = \frac{\partial T}{\partial x}\cdot\frac{\partial x}{\partial r} + \frac{\partial T}{\partial y}\cdot\frac{\partial y}{\partial r} \qquad \frac{\partial\bar{T}}{\partial\theta} = \frac{\partial T}{\partial x}\cdot\frac{\partial x}{\partial\theta} + \frac{\partial T}{\partial y}\cdot\frac{\partial y}{\partial\theta}$$

$$\frac{\partial\bar{T}}{\partial r} = (3x^2 - 3y^2)\frac{\partial x}{\partial r} - 6xy\frac{\partial y}{\partial r}$$

$$\begin{aligned}&= 3r^2(\cos^2\theta - \sin^2\theta)(\cos\theta) - 6r^2\cos\theta\sin\theta(\sin\theta)\\ &= 3r^2(\cos^3\theta - 3\cos\theta\sin^2\theta) = 3r^2\cos 3\theta.\end{aligned}$$

$$\frac{\partial\bar{T}}{\partial\theta} = (3x^2 - 3y^2)\frac{\partial x}{\partial\theta} - 6xy\frac{\partial y}{\partial\theta}$$

$$\begin{aligned}&= 3r^2(\cos^2\theta - \sin^2\theta)(-r\sin\theta) - 6r^2\cos\theta\sin\theta(r\cos\theta)\\ &= 3r^3(\sin^3\theta - 3\sin\theta\cos^2\theta) = -3r^3\sin 3\theta.\end{aligned}$$

If we instead begin with $\bar{T} = r^3\cos 3\theta$, we obtain

$$\begin{aligned}\frac{\partial\bar{T}}{\partial r} &= 3r^2\cos 3\theta,\\ \frac{\partial\bar{T}}{\partial\theta} &= -3r^3\sin 3\theta.\end{aligned}$$

9. (a) $\dfrac{\partial}{\partial x} = \dfrac{\partial}{\partial u}\dfrac{\partial u}{\partial x} + \dfrac{\partial}{\partial v}\dfrac{\partial v}{\partial x} = \dfrac{\partial}{\partial u} + \dfrac{\partial}{\partial v} \qquad \dfrac{\partial}{\partial y} = \dfrac{\partial}{\partial u}\dfrac{\partial u}{\partial y} + \dfrac{\partial}{\partial v}\dfrac{\partial v}{\partial y} = -4y\dfrac{\partial}{\partial u} + 4y\dfrac{\partial}{\partial v}$

$$\text{since} \quad \frac{\partial u}{\partial x} = 1, \quad \frac{\partial v}{\partial x} = 1, \quad \frac{\partial u}{\partial y} = -4y, \quad \frac{\partial v}{\partial y} = 4y.$$

(b) $\dfrac{1}{4y}\dfrac{\partial}{\partial y} = -\dfrac{\partial}{\partial u} + \dfrac{\partial}{\partial v}$ so

$$\frac{1}{4y}\frac{\partial f}{\partial y} + \frac{\partial f}{\partial x} = -\frac{\partial f}{\partial u} + \frac{\partial f}{\partial v} + \frac{\partial f}{\partial u} + \frac{\partial f}{\partial v} = 2\frac{\partial f}{\partial v} = f.$$

(c) $2f_v = f$ is a partial differential equation for $f$. Consider the separable ordinary differential equation

$$2\frac{\mathrm{d}f}{\mathrm{d}v} = f.$$

This is a separable ODE, which we rearrange to obtain

$$\begin{aligned}&\int\frac{1}{f}\,\mathrm{d}f = \frac{1}{2}\int\,\mathrm{d}v,\\ \Rightarrow &\ln f = \frac{v}{2} + c,\\ \Rightarrow &f = A\mathrm{e}^{v/2},\end{aligned}$$

where $A = \mathrm{e}^c$ is an arbitrary constant. Hence, the PDE $f_v = v$ has the solution $f = A\exp\left(\frac{1}{2}v\right)$, with arbitrary "constant" $A$ being a function of $u$. hence $f = A(u)\exp\left(\frac{1}{2}v\right)$, and

$$f(x,y) = A(x-2y^2)\exp\left(\frac{1}{2}\left(x+2y^2\right)\right),$$

for some function $A(\cdot)$.

10. To find stationary points of $f(x,y) = (x^2+y^2)^2 + 2a^2(y^2-x^2)$, we must solve

$$\begin{aligned} f_x &= 4x(x^2+y^2) - 4a^2x = 0 \Rightarrow x(x^2+y^2-a^2) = 0, \\ f_y &= 4y(x^2+y^2) + 4a^2y = 0 \Rightarrow y(x^2+y^2+a^2) = 0. \end{aligned}$$

One solution to the system is $x = y = 0$, i.e. $(0,0)$ is a stationary point. To find the remaining stationary points, set $y = 0$ then the second equation is satisfied and $x$ satisfies $x^2 = a^2 \Rightarrow x = \pm a$. Note that $x = 0$ yields no further real solutions, since $y$ must satisfy $y^2 = -a^2$. Hence, $f$ has stationary points at $(0,0)$ and $(\pm a, 0)$.

To determine the nature of the stationary points, we find $f_{xx} = (4+8x^2)(x^2+y^2) - 4a^2$, $f_{yy} = (4+8y^2)(x^2+y^2) + 4a^2$, $f_{xy} = 8xy(x^2+y^2)$.

$\underline{(x,y) = (0,0)}$

$f_{xx}f_{yy} - f_{xy}^2 = -16a^2 < 0$, so $(0,0)$ is a saddle point.

$\underline{(x,y) = (\pm a, 0)}$

$f_{xx}f_{yy} - f_{xy}^2 = 4(4+8a^2)a^2 > 0$, $f_{xx} = (4+8a^2)a^2 > 0$, so $(\pm a, 0)$ are both local minima.